Perfect Matching Is Hard

What they did and why they did so; and a little bit of what they *did not* do!

10th May 2024

The Research Program

Cook And Reckhow, 73

Given an propositional formula $\phi \in \mathtt{UnSAT}$, give me a proof π of the claim.

1. If $|\pi| = \mathtt{poly}(|\phi|)$ then we have $\mathtt{CoNP} = \mathtt{NP}$

2. If we have a lower bound showing that $|\pi| = \omega(\mathtt{poly}(|\phi|)$ then $\mathtt{NP} \neq \mathtt{CoNP}$

Widely conjectured that **(2)** is true, but we're nowhere close to proving this

No restriction on what a proof looks like. Just a an arbitrary sequence of words that some algorithm called the verifier can process.

Example Of A Proof System

Truth Table

- Given a propositional formula ϕ, the proof π is just the entire truth table of ϕ
- Verification involves processing each row of the table and checking that no labelling of the inputs evaluates to 1.
- This is a very restricted language for providing proofs.
- Certainly not efficient. Quite easy to see that if we just used truth tables as the language for proofs

Modified Research Program

Prove lower bounds for a more restricted class of proof systems

- **A dynamic proof system:** A proof is an ordered sequence of lines $F_1, \ldots, F_t$ called proof lines
- We'll get to what the lines look like in a bit.
- A proof line can either be an *axiom* or it can be derived from a *set of inference rules*
- The axioms and inference rules describe a language for describing the proofs

A Slightly Richer Language For Proofs

Resolution

- We are given some CNF formula $\phi = C_1 \wedge \ldots C_m$ that is unsatisfiable (refutation is ϕ is tautology)
- The proof lines are clauses **(disjunction of literals)**
- Has one inference rule : $\dfrac{C \vee x \qquad D \vee \neg x}{C \vee D}$
- If you want to prove a claim then you start with axioms, then you get to the last line of the proof which will be ϕ
- If you want to refute a claim, you start with ϕ and end with some contradiction like $1 = 0$

We've seen two proof systems, are there more?

Yes. Many many more!

What do these arrows mean p-domination?

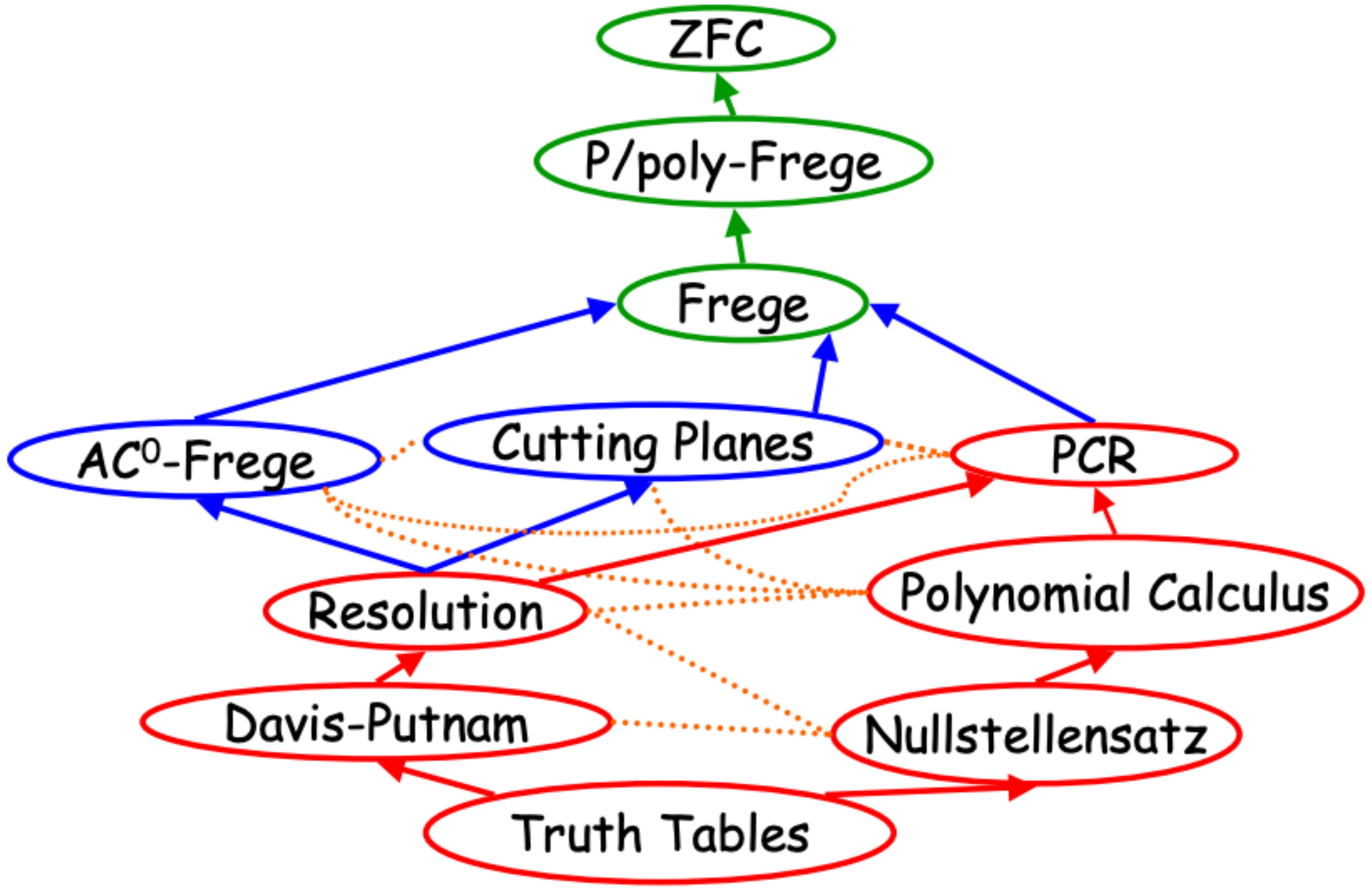

Figure 8. Some proof system relationships

[Figure Source: https://www.cs.cornell.edu/~sabhar/publications/iaspcmi-proofcomplexity00.pdf](https://www.cs.cornell.edu/~sabhar/publications/iaspcmi-proofcomplexity00.pdf)

Which Proof Systems Do We Care About?

Sum of squares and Polynomial Calculus

- Instead of a propositional formula ϕ we get a set of polynomial constraints
- Proofs are polynomials as well.
- First we describe a simpler proof system: Nullstellensatz proof systems

Click to expand notes for more details

Polynomial Calculus (PC)

Nullstellensatz Proof System (Pre-view to PC)

- Instead of ϕ we get a polynomial constraints
- Example: $\phi(x, y, z) = x \vee \neg y \vee z$ can be expressed with polynomial constraints $p(x, y, z) = 0; f_1(x, y, z) = 0, f_2(x, y, z) = 0, f_3(x, y, z) = 0$ over some field $\mathbb{F}$
- $f_1 := x(1 - x) = 0, f_2 := y(1 - y) = 0, f_3 := z(1 - z) = 0$
- $p := (1 - x)(y)(1 - z) = 0$
- So now the question is asking given polynomial constraints p, f_1, f_2, f_3 show that there exists no labelling of x, y, z in the algebraic closure of $\mathbb{F}$ such that these constraints are satisfied.

Polynomial Calculus (PC)

Nullstellensatz Proof System (Pre-view to PC)

Theorem:

Fix $n, m \in \mathbb{N}$. Let $\mathbb{F}$ be a field and $f_1, \ldots, f_m$ be multivariate polynomials in $\mathbb{F}[x_1, \ldots, x_n]$ (polynomials with coefficients in $\mathbb{F}$). Then the system of equations

$$\begin{aligned} f_1(x_1, \ldots, x_n) &= 0 \\ f_2(x_1, \ldots, x_n) &= 0 \\ &\cdots \\ f_m(x_1, \ldots, x_n) &= 0 \end{aligned}$$

has **no solution** in the algebraic closure of $\mathbb{F}$ **if and only if** there exists polynomials $g_1, \ldots, g_m \in \mathbb{F}[x_1, \ldots, x_n]$ such that $\sum_{i=1}^{m} g_i f_i \equiv 1$

Polynomial Calculus (PC)

continuing from Nullstellensatz

- Ok we have a language for expressing proofs for unsatisfiability — you give me polynomials, i give you polynomials as a proof.
- How do you measure proof size?
- How many bits do you need for a compact representation of a polynomial ?
- The answer depends on the degree of the polynomial (coefficient representation)
- So the complexity of this proof system can be described by the maximum degree of any polynomial the verifier must work with. For the proof system described above it is $\max_{j=1\ldots,m} g_i f_i$

Polynomial Calculus (PC)

continuing from Nullstellensatz

- But the proof system described above has some limitations; it's not really dynamic.
- You give me a proof $\pi = (g_1, \ldots, g_m)$ for problem $f_1, \ldots, f_m$ and the verifier just computes $\langle \vec{g}, \vec{f} \rangle$
- Polynomial calculus is the dynamic version of this proof system, where you can start with $\vec{f}$, use $\vec{g}$ and write down more proofs based on some inference rules.
- So what are the rules?
- (Linear combination) $f_j = a \cdot f_i + b \cdot f_{i'} = 0$ for some $a, b \in \mathbb{F}$, and $0 \leq i, i' < j$ or
- (Multiplication by variable) $f_j = x_k \cdot f_i$ for $i < j$ and $k \in [n]$.

Sum Of Squares (SOS)

Semi-algebraic Proof Systems

- The proof system described above had the language of polynomials but the constraints for always equalities.
- Now we will allow for inequalities in the constraints as well.
- Furthermore, we will restrict our field to the reals only $\mathbb{R}$ **(not any arbitrary field)**.
- With these constraints, we do not actually get to SOS proof systems, but we get to its precursor the Sherali Adams proof system.
- We describe that first, modify it to get SOS

Sherali Adams

Semi-Algebraic Proof Systems

Sherali-Adams Proofs (Definition)

Given a system of polynomial inequalities and equalities over the reals denoted by $F = (h_1 \geq 0, \ldots, h_m \geq 0; f_1 = 0, \ldots, f_{m'} = 0)$ with variables $x_1 \ldots, x_n$, a Sherali-Adams derivation of a polynomial inequality $h \geq 0$ (**the final decision**) is an explicit list of real polynomials $\pi = (g_0, \ldots, g_m; e_1, \ldots, e_{m'+n})$ such that

$$h = g_0 + \sum_{i=1}^{m} g_i h_i + \sum_{i=1}^{m'} e_i f_i + \sum_{i=m'+1}^{n} e_i(x_i^2 - x_i)$$

where each g_i is a positive linear combination of generalized monomials of the form $g_i = \sum_{P,N \subseteq [n]} c_{(P,N)} \prod_{i \in P} x_i \prod_{i \in N} (1 - x_i)$ for disjoint subsets $P, N \subseteq [n]$ (these are negated and positive literals from the clause), and $c_{(P,N)} \in \mathbb{R}$. For the equalities, we have no constraint on the form of their coefficient polynomials.

Sum Of Squares

Instead of generalised monomials get sum of squares polynomials

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where each g_i is a positive linear combination of generalized monomials of the form [redacted] for disjoint subsets $P, N \subseteq [n]$ (these are negated and positive literals from the clause), and $c_{(P,N)} \in \mathbb{R}$. For the equalities, we have no constraint on the form of their coefficient polynomials.

$$g_i = \sum_{j=1}^{r_i} p_{i,j}^2$$

$p_{i,j} \in \mathbb{R}[x_1, \ldots, x_n]$ in standard monomial form

Sum Of Squares

Instead of generalised monomials get sum of squares polynomials

SOS Proofs (Definition)

Given a system of polynomial inequalities and equalities over the reals denoted by $F = (h_1 \geq 0, \ldots, h_m \geq 0; f_1 = 0, \ldots, f_{m'} = 0)$ with variables $x_1 \ldots, x_n$, a Sherali-Adams derivation of a polynomial inequality $h \geq 0$ (**the final decision**) is an explicit list of real polynomials $\pi = (g_0, \ldots, g_m; e_1, \ldots, e_{m'+n})$ such that

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where each g_i is sum-of-squares (sos) i.e. $g_i = \sum_{s=1}^{r_i} p_{i,s}^2$ for $p_{i,s} \in \mathbb{R}[x_1, \ldots, x_n]$ (in standard monomial form). For the equalities, we have no constraint on the form of their coefficient polynomials.

Ok! We now have proof systems in the language of polynomials, but what on earth does it have to do with Perfect Matching or even graphs??

Some basic Notation On Graphs

Matchings in Graphs

Converting statements about graphs to statements about polynomials

Graphs $\to$ Polynomial Constraints

Define variables $x_1, \ldots, x_{|E|}$ over the reals. The following constraint restricts each $x_e \in \{0,1\}$ for all $e \in E$.

$$x_e(1 - x_e) = 0$$

Let $\vec{b} = (b_1, \ldots, b_{|V|}) \in \mathbb{Z}^{|V|}$ be a constant vector. Then, for each $v \in V$, define the polynomials constraints f_v for all $v \in V$ as:

$$f_v := \sum_{e \in E} x_e - b_v = 0$$

In the paper, they define $\texttt{Card}(G, \vec{b})$ as this system of linear equations along with the Boolean constraints. Note that $\texttt{Card}(G, 1^{|V|})$) corresponds to the perfect matching.

Look, I have a set of polynomials that describe some unsatisfiable language — this graph has **NO** perfect matching.

Find me a proof of this property i.e for no assignment of the variables $x_1, \ldots, x_{|E|}$ are the constraints satisfied.

Main Result

THEOREM 1.1. *There is a constant d_0 such that for all constants $d \geq d_0$ and $t \in [d]$, the following holds asymptotically almost surely over a random d-regular graph G on n vertices.*

1. $\mathsf{Deg}(\mathsf{Card}(G,\vec{t}) \vdash_{PC_{\mathbb{F}}} \bot) = \Omega(n/\log n)$ *for any fixed field* $\mathbb{F}$ *with* $\mathrm{char}(\mathbb{F}) \neq 2$.
2. $\mathsf{Deg}(\mathsf{Card}(G,\vec{t}) \vdash_{SoS} \bot) = \Omega(n/\log n)$.
3. *There is a* $\delta > 0$ *such that* $\mathsf{Size}(\mathsf{Card}(G,\vec{t}) \vdash_{\mathscr{F}_D} \bot) = \exp\left(\Omega(n^{\delta/D})\right)$, *for all* $D \leq \frac{\delta \log n}{\log\log n}$.

Attention Rajko

I have no idea yet if this lower bound is bad or not — or a reference to why this is an interesting result

How do they show this? Next time...

What about the other two theorems in their paper?

The second theorem

- The second theorem helps prove the first theorem, and has to do with embeddings (we will get there next time)

- Additionally, it also helps resolve some other questions from Filmus.

The second theorem

- These theorem is proved in order to prove the first theorem, and has to do with embeddings.

This results is really a means to prove the first result. Paraphrasing their proof strategy from the paper, we get that the way they prove hardness results is:

> **Proof Idea**
>
> - Start $\Omega(n)$ worst case degree lower bound for sparse graphs. **Question**: There is already an existing lower bound for sparse graphs for the perfect matching problem in PC by Buss, Pitassi, Impagliazzo, Grigoriev.
> - Then the task is coming up with a larger host graph G that is d regular (and also an α-expander) but contains the above sparse graph as a subgraph. We need to construct G in such a way that the original hardness of the problem H is not lost.